

Safestep

SafeStep is an autonomous intervention device that is aimed at directly mitigating the effects of a fall in an inpatient setting. This device was made in collaboration with Baystate Medical Center and features a mechanical classification detection system and a CO₂ inflation valve that deploys an airbag.

The prevention of injurious falls among hospitalized patients is a common challenge, as almost 1 million patients annually experience a fall leading to some form of injury. These falls can occur quite sporadically, as many patients are unaware of their muscle fatigue when standing up. Primarily, Baystate Medical Center has an elderly inpatient setting where falls can cause severe injury. The objective of this device is to prevent falls directly, thus reducing any forms of injury and extended hospital costs. These falls can lead to severe complications such as fractures, head injuries, and prolonged hospital stays.

Patients who are most at risk of falling in hospitals are elderly patients 60 and older, along with patients who are 50 years old with two or more comorbidities. Hospital patients with these falls are typically aged 60 and older and fall due to age-related mobility decline, lack of stability, and unwillingness to request assistance from nurses.

SafeStep is engineered as an untethered wearable device integrated seamlessly into a patient's hospital gown (Johnny). The system device is driven by an Adafruit Feather M4 Express, housing a 32-bit ARM Cortex-M4 microcontroller (ATSAMD51) operating with 512 KB of Flash memory. The sensor used was an MPU-6050 6-axis Inertial Measurement Unit (IMU) consisting of an integrated 3-axis accelerometer and a 3-axis gyroscope communicating through an I²C bus. The microcontroller runs a trained Random Forest machine learning model optimized to distinguish true fall signatures from daily activities such as lying down, standing up, or sitting. The system is powered by a 3.7V LiPo Battery and a separate 12V IRLB8721 N-Channel MOSFET powering an electromechanical solenoid valve release. Upon a confirmed classification from the machine learning model, the solenoid release valve activates, exposing CO₂ gas into an 8.1 L airbag in under 200 ms.

The primary failure mode in elderly falls is a hip fracture, which occurs when the impact force exceeds the mechanical strength of the femur. Research indicates that the

average peak force required to fracture an elderly femur is 2984 ± 913 N. The equation from the Impulse-Momentum Theorem $F \times \Delta t = \Delta p$

Without protection, a fall on a hard hospital floor results in a near-instantaneous impact (very small t), causing the peak force (F) to spike well above the 3000 N fracture threshold. SafeStep deploys gas-filled pads that compress upon impact. This compression significantly increases the time of impact (t). Since the momentum change (p) is constant for a given fall height, increasing t quantitatively drives the peak force (F) down. The group's analysis below confirms that by extending the impact duration to > 130 ms (meaning the inflated pads cushion the fall and increase the stopping time to at least 0.13 seconds), the system reduces the transmitted force to < 2000 N, providing a safety factor that keeps the impact well below the fracture threshold.